

Introduction to Artificial Intelligence

What is Artificial Intelligence?

Artificial Intelligence (AI) is a field of computer science focused on creating systems that can perform tasks that normally require human intelligence. These tasks include learning, reasoning, understanding language, recognizing images, and making decisions.

Machine Learning

Machine Learning (ML) is a branch of AI in which computers learn patterns from data instead of being explicitly programmed for every task. Common types of machine learning include supervised learning, unsupervised learning, and reinforcement learning.

Deep Learning

Deep Learning is a subset of machine learning that uses neural networks with multiple layers. Deep learning is widely used for image recognition, speech recognition, natural language processing, and recommendation systems.

Natural Language Processing

Natural Language Processing (NLP) enables computers to process and understand human language. Applications include chatbots, translation, sentiment analysis, text summarization, and question answering.

Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) combines information retrieval with a language model. A RAG system first retrieves relevant information from a knowledge base and then provides that information to a language model to generate a grounded answer. RAG can reduce the need for a model to rely only on information stored in its parameters.

Embeddings

Embeddings are numerical representations of text, images, or other data. Similar pieces of information have embeddings that are close to each other in vector space. RAG systems commonly use text embeddings to perform semantic similarity search.

Vector Databases

A vector database stores embeddings and supports similarity searches. FAISS is a popular library for efficient similarity search over dense vectors. In a PDF question answering system, document chunks can be converted into embeddings and stored in FAISS.

Example RAG Workflow

A basic PDF RAG workflow is: load the PDF, extract text, split the text into chunks, create embeddings, store the embeddings in a vector database, retrieve relevant chunks for a user question, and send the retrieved context to a language model.

Example Questions

You can ask questions such as: What is artificial intelligence? What is machine learning? How does RAG work? What are embeddings? What is FAISS? What is the difference between AI and machine learning?